African smart cities: A critical analysis

An exercise in open source intelligence using an AI-powered search engine.

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Introduction

It may be strange to start a paper with its methodology, but in this case it is necessary. This entire study was researched and written by Perplexity, an AI-powered answer engine that combines large language models with real-time internet search capabilities. Unlike traditional search engines that return lists of links, Perplexity interprets natural language questions, conducts comprehensive web searches across multiple sources and, most importantly, generates results with inline citations linking to original sources. This transparency in sourcing enables relatively easy verification of information to maintain research rigour. For this study, Perplexity was employed to systematically gather, synthesise, and cross-reference academic and official policy documents along with news from national and trade press sources, significantly accelerating the literature review process. My role in this, as in all my open source intelligence projects, is to design the concept, build metadata, write instructions and review outcomes. **The full methodology, outlining the design of the project through incremental iterative interactions with Perplexity, can be found here.**

African smart city initiatives examined encompass a broad range of technological deployments and digital transformation efforts. These include intelligent transport systems designed to alleviate chronic traffic congestion, extensive closed-circuit television surveillance networks marketed under "Safe City" frameworks, digital service delivery platforms for government services and revenue collection, smart water and waste management systems, digital identity programs, and ambitious greenfield developments such as Egypt's New Administrative Capital and Kenya's Konza Technopolis. The initiatives also extend to sector-specific applications including digital health systems, e-education platforms, smart agriculture solutions, and fintech innovations that have achieved varying degrees of market penetration.

This analysis examines initiatives across twelve major African urban centres: Abidjan, Accra, Cairo, Dakar, Johannesburg, Kampala, Kigali, Lagos, Lusaka, Nairobi, Rabat, and Tunis. The selection, based on rankings in the African Smart City Index 2024, reflects a spectrum of development levels, from emerging technology hubs like Nairobi and Kigali to established metropolitan centres such as Johannesburg and Cairo, alongside rapidly urbanizing cities like Lagos and Accra.

Common features emerge across these diverse contexts. Nearly all twelve cities have deployed or are planning surveillance systems with facial recognition capabilities, frequently supplied by Chinese technology firms. Intelligent transport systems feature prominently in Nairobi, Johannesburg, Cairo, and Rabat. Digital government platforms have been implemented across most cities, with varying degrees of comprehensiveness and citizen uptake. Mobile money systems have achieved remarkable success in East Africa, with Kenya's M-Pesa serving as a continental model. Greenfield smart city developments are also emerging, from Egypt's New Administrative Capital to Senegal's Diamniadio Lake City and Morocco's Mohammed VI Technopolis.

The rationale for selecting these twelve cities rests on their collective ability to illuminate continental trends while representing the diversity of African urban experiences. They include capital cities serving as administrative and political centres, commercial hubs driving economic activity, and technology startup ecosystems attracting venture capital. They represent different colonial legacies, governance systems, and levels of democratic consolidation. This diversity enables analysis of how smart city technologies interact with varying institutional contexts, regulatory frameworks, and civil society capabilities. The cities also reflect different stages of smart city development, from early pilot programs to mature deployments, allowing examination of implementation challenges, adaptation processes, and long-term sustainability questions.

Despite significant concerns, smart city initiatives across these twelve African cities have generated tangible benefits that deserve recognition and are addressed in the first three sections. These benefits manifest across social, economic, and environmental dimensions, though their distribution remains uneven and their sustainability uncertain.

Notwithstanding these benefits the analysis that follows of governance, financing, ownership, inclusivity, accountability and surveillance reveals profound concerns

that undermine the transformative potential of digital urban development and pose significant risks to rights, equity, and democratic governance.

Social Benefits

Digital government platforms have demonstrably improved access to public services in several cities. Nairobi's eCitizen portal provides access to over 16,000 government services from more than 100 agencies, significantly reducing the time and cost citizens previously expended traveling to government offices and navigating bureaucratic processes. Kigali's Irembo platform has achieved remarkable uptake with over 3 million registered users accessing services including birth certificates, business registration, and driving licenses through digital channels. These platforms have reduced processing times from weeks to days or hours, decreased opportunities for petty corruption by limiting face-to-face interactions with officials and provided transparent tracking of application status.

Revenue collection has improved markedly through digital systems. Accra's digitalization of property taxes and business operating permits has increased municipal revenue collection rates, providing resources for service delivery. Lagos has enhanced tax compliance through digital platforms that simplify payment processes while improving transparency. Nairobi County's digital revenue collection system has reduced leakage and improved fiscal capacity. These improvements, while benefiting government budgets, also create more predictable environments for businesses and reduce arbitrary enforcement.

Water service delivery has been enhanced through smart metering and management systems in multiple cities. Nairobi's deployment of IoT-enabled smart water meters has improved billing accuracy, reduced non-revenue water losses through leak detection, and provided consumers with real-time consumption data enabling conservation behaviours. The Maji Wazi project in Nairobi's Mathare informal settlement demonstrated that appropriate technology deployed with community participation can extend services to previously underserved areas while building local capacity through training resident field engineers. Kigali's smart water management system has improved distribution efficiency in a city facing water scarcity challenges.

Healthcare access has expanded through telemedicine platforms and digital health records in several cities. Kigali's integration of electronic medical records across health facilities has improved continuity of care and reduced duplication of diagnostic tests. Mobile health applications have extended health information to rural areas surrounding urban centres. Digital appointment booking systems have reduced waiting times and improved patient flow in overcrowded public hospitals. During the COVID-19 pandemic, digital contact tracing systems and vaccination registration platforms demonstrated the potential of digital infrastructure to respond to public health emergencies, though concerns about data retention and purpose limitation emerged.

Education access has been enhanced through e-learning platforms, particularly following COVID-19 disruptions. Several cities deployed digital learning solutions that enabled education continuity when physical schools closed. While digital divides limited reach, these initiatives demonstrated potential for supplementing traditional education delivery. Digital libraries and online resources have expanded access to educational materials beyond physical textbook availability. Skills training platforms have emerged in cities like Nairobi, Lagos, and Johannesburg, connecting youth to digital economy opportunities, though scalability remains limited.

Financial inclusion has advanced dramatically through mobile money systems, particularly in East Africa. Kenya's M-Pesa has become the continental exemplar, providing financial services to previously unbanked populations and enabling digital economic participation. The system processed billions of dollars in transactions in recent years, serving as infrastructure for salary payments, bill payments, savings products, and credit access. Small merchants previously operating purely in cash have gained access to digital payment ecosystems. Similar systems in other East African cities have replicated aspects of this success, though West African cities have experienced lower penetration rates due to regulatory barriers, lower mobile penetration, and more fragmented telecommunications markets.

Transportation efficiency has improved in cities deploying intelligent transport systems, though benefits remain concentrated in affluent corridors. Nairobi's Intelligent Transport System, once fully operational, promises to reduce journey times through optimized traffic signal coordination and real-time traffic

management. Cairo's Intelligent Transportation System has improved traffic flow on major arteries, reducing congestion and air pollution from idling vehicles. Johannesburg's Integrated Rapid Public Transit Network, supported by digital ticketing and fleet management systems, has improved public transportation reliability, though spatial coverage remains limited. Digital ride-hailing platforms operating in all twelve cities have expanded transportation options and created income opportunities for drivers, though concerns about labour rights and fair compensation persist.

Economic Benefits

Job creation through the digital economy represents a significant economic benefit, though numbers remain modest relative to unemployment challenges. Nairobi's technology startup ecosystem secured $638 million in venture capital funding in 2024, supporting thousands of direct jobs in technology companies and many more indirect positions in supporting services. Lagos has emerged as West Africa's technology hub, with the Yaba technology cluster attracting investment and generating employment for software developers, digital marketers, and technology entrepreneurs. Kigali's positioning as a technology services hub has attracted business process outsourcing operations creating formal sector jobs. Johannesburg continues to dominate South Africa's technology sector, with established companies and startups providing employment opportunities.

Entrepreneurship enablement through digital platforms has created opportunities for small businesses and individual entrepreneurs. Digital marketplace platforms connect producers to consumers, reducing traditional middleman costs. Mobile money systems have enabled micro-enterprises to operate more efficiently through digital payments. Digital lending platforms, utilizing alternative credit scoring based on mobile money transaction histories and other data sources, have extended credit access to small businesses previously excluded from formal banking. While concerns about predatory lending practices and high interest rates persist, these innovations have filled credit gaps that traditional banks neglected.

Trade facilitation through digital platforms and improved connectivity has reduced transaction costs for businesses engaged in cross-border commerce. Single-window systems for customs clearance, implemented with varying degrees of success, have reduced clearance times and opportunities for corruption. Digital documentation systems have simplified compliance requirements. Regional

integration initiatives supported by digital infrastructure have potential to facilitate intra-African trade, though implementation remains incomplete and non-tariff barriers persist.

Agricultural productivity improvements through digital agriculture solutions have benefited some farmers, though scalability remains limited. Digital extension services provide agronomic advice via mobile platforms, reaching farmers previously unserved by extension agents. Weather information services enable better planning for planting and harvesting. Digital marketplace platforms connect farmers to buyers, potentially improving price transparency and reducing post-harvest losses. Precision agriculture technologies, while limited to larger commercial farms, demonstrate potential for productivity improvements. Mobile money has facilitated input financing and output payment systems, improving agricultural value chain efficiency.

Tourism promotion through digital platforms and improved connectivity has enhanced cities' abilities to attract visitors and showcase cultural assets. Digital booking platforms, online payment systems, and improved digital connectivity have reduced friction in tourism transactions. Smart city infrastructure, particularly in cities like Kigali and Rabat, has enhanced destination attractiveness. Digital cultural heritage platforms have promoted local heritage while generating income opportunities for cultural entrepreneurs. Tourism-related employment has benefited from these digital enablers, though the COVID-19 pandemic demonstrated sector vulnerability.

Innovation ecosystem development has progressed in several cities, creating foundations for knowledge economy participation. Technology hubs, incubators, and accelerators have emerged in Nairobi, Lagos, Johannesburg, Accra, Cairo, and other cities, providing infrastructure, mentorship, and networking opportunities for entrepreneurs. University-industry linkages have strengthened in some contexts, though generally remain weak relative to developed economies. Digital skills training programs have expanded, though quality varies widely and alignment with market demands remains imperfect. Government support for innovation, through technology parks and incentive programs, has increased, though implementation often falls short of policy rhetoric.

Environmental Benefits

Smart waste management systems have improved collection efficiency and reduced environmental impacts in several cities. Kigali's integrated solid waste management system, supported by digital tracking and route optimization, has maintained the city's reputation for cleanliness while improving collection coverage. Konza Technopolis in Kenya has implemented Africa's first pneumatic waste collection system, demonstrating potential for innovative approaches in new developments. Digital waste tracking systems in several cities have improved regulatory oversight and reduced illegal dumping, though enforcement capacity remains limited. Waste-to-energy initiatives, while limited in scale, demonstrate potential for circular economy approaches enabled by digital monitoring and optimization systems.

Energy efficiency improvements through smart grid technologies and building management systems have reduced consumption and emissions in some contexts. Smart metering enables time-of-use pricing and provides consumers with consumption data to inform conservation behaviours. Building management systems in commercial and government buildings have optimized heating, cooling, and lighting, reducing energy waste. Renewable energy integration has been facilitated by smart grid technologies that manage intermittent supply from solar and wind sources. Johannesburg's smart grid pilot demonstrated potential for improving distribution efficiency and integrating distributed generation, though citywide deployment faces financing constraints.

Air quality monitoring through sensor networks has improved environmental awareness and provided data for policy interventions. Several cities have deployed air quality monitoring systems providing real-time data to citizens and policymakers. This information has raised public awareness about pollution sources and health impacts, creating pressure for regulatory action. Data-driven identification of pollution hotspots has enabled targeted interventions. While monitoring does not directly reduce pollution, the information foundation it provides is essential for evidence-based environmental management.

Water resource management has benefited from digital monitoring and optimization systems. Smart irrigation systems have improved water use efficiency in urban agriculture and landscaping. Leak detection systems have reduced non-revenue water losses, extending limited water resources. Water quality monitoring systems have improved public health protection through early warning of

contamination events. Integrated water resource management systems have enhanced coordination across watersheds, though inter-jurisdictional cooperation challenges persist.

Climate adaptation planning has been informed by digital data collection and modelling systems. Climate monitoring networks provide data on local climate trends and extreme weather events. Digital modelling tools inform infrastructure planning and climate risk assessment. Early warning systems for floods and other climate-related hazards have improved community preparedness and emergency response. Green building standards in new developments like Egypt's New Administrative Capital incorporate environmental considerations from design stages, though implementation monitoring remains limited.

Transportation emissions reductions from intelligent transport systems and public transit digitalization have yielded environmental benefits. Optimized traffic flow reduces emissions from idling vehicles. Improved public transportation reliability encourages modal shift from private vehicles. Electric vehicle charging infrastructure, deployed in limited quantities in cities like Kigali and Johannesburg, supports clean transportation transitions. Digital ride-sharing platforms have potential to reduce vehicle miles travelled through improved utilization, though evidence suggests they may increase congestion by competing with public transit.

Governance

Governance of smart city initiatives across the twelve cities exhibits troubling patterns of institutional fragmentation, unclear coordination mechanisms, policy incoherence, and capacity deficits that undermine effective implementation and democratic oversight. Rather than integrated smart city strategies with clear lines of authority and accountability, most cities display archipelagos of disconnected projects operating under different institutional arrangements with minimal coordination.

The institutional landscape in Nairobi exemplifies this fragmentation. The county government's Innovation and Digital Economy Sector manages some initiatives, the national government's Konza Technopolis Development Authority pursues separate projects, the Kenya Urban Roads Authority implements transport systems, and the National Police Service operates surveillance networks. No overarching coordination mechanism ensures these initiatives work

synergistically or avoid duplication. Similar fragmentation appears in Johannesburg, where metropolitan initiatives operate alongside provincial and national programs with overlapping mandates and unclear demarcation of responsibilities. Cairo's smart city governance spans multiple ministries, the New Urban Communities Authority, and the Administrative Capital for Urban Development company, creating a complex institutional maze resistant to coordination.

This fragmentation has predictable consequences. Resources are wasted on duplicative systems and incompatible technologies. Citizens face disconnected digital interfaces requiring separate registrations and authentication processes for different services. Data cannot flow across systems to enable the integrated service delivery that smart city rhetoric promises. Accountability becomes diffuse as no single entity bears responsibility for overall outcomes. Private vendors exploit coordination gaps to lock different agencies into incompatible proprietary systems, increasing long-term costs and dependencies.

Policy frameworks compound governance challenges through incoherence and gaps between rhetoric and implementation. Many cities have adopted smart city strategies with ambitious vision statements but limited concrete implementation plans, resource allocations, or accountability mechanisms. Rabat's smart city strategy, Lagos's Smart City Blueprint, and Accra's digital transformation plans exemplify aspirational documents that guide actual implementation only loosely. National digital transformation strategies, where they exist, often lack mechanisms for coordinating with municipal initiatives, creating parallel tracks that compete for resources and attention.

Regulatory frameworks lag technological deployments, creating governance vacuums where smart city technologies operate with limited oversight. Data protection regulations, where enacted, frequently lack implementing regulations, resourced enforcement agencies, or penalty provisions that deter violations. Procurement regulations designed for traditional infrastructure prove inadequate for rapidly evolving digital technologies, creating opportunities for corruption and unfavourable contract terms. Public-private partnership frameworks lack specificity regarding data ownership, intellectual property rights, and technology transfer requirements, resulting in contracts heavily favouring vendors.

Capacity deficits within government institutions severely constrain effective governance. Technical capacity to evaluate technology proposals, negotiate favourable contract terms, and manage complex implementations remains limited. Many cities lack in-house technical expertise, creating asymmetric information relationships with vendors and consultants. When governments hire external consultants, conflicts of interest often emerge as consultants maintain vendor relationships. Political leadership frequently lacks digital literacy, making evidence-based decision-making difficult and rendering leaders vulnerable to vendor capture through aggressive marketing and demonstration projects.

Political economy dimensions further undermine governance quality. Smart city initiatives become political projects serving legitimation functions for authoritarian regimes or electoral vehicles for democratic politicians rather than development interventions subject to rigorous evaluation. Egypt's New Administrative Capital serves regime consolidation objectives as much as administrative efficiency goals. Rwanda's smart city positioning reinforces the government's modernization narrative while constraining critical examination of outcomes and costs. Procurement processes become sites of corruption and patronage distribution rather than value-for-money optimization. Contract awards favour politically connected firms over competitive bidding. Oversight institutions face political pressure to avoid embarrassing audits or investigations.

Multi-stakeholder governance, emphasized in continental frameworks like the AU Digital Transformation Strategy, remains largely rhetorical. Civil society participation in smart city planning and oversight proves superficial, limited to consultation ceremonies rather than genuine co-design or shared decision-making authority. Private sector engagement concentrates among large telecommunications firms and international technology vendors, excluding small local technology companies. Academic institutions remain peripheral to policy processes despite potential research contributions. Marginalized communities most affected by smart city deployments, including informal settlement residents facing surveillance and digital service delivery requirements, have minimal voice in governance processes.

Coordination across scales—municipal, metropolitan, provincial, national, regional—proves elusive despite rhetoric about integrated governance. Continental frameworks like the AU Digital Transformation Strategy and national

digital strategies fail to effectively guide municipal implementations. Regional Economic Communities have developed policy frameworks, but member state compliance and implementation remain voluntary and uneven. Cities pursuing smart city initiatives operate largely autonomously, missing opportunities for shared learning, joint procurement, and standards harmonization that could reduce costs and improve outcomes.

Temporal governance challenges receive insufficient attention. Smart city technologies require ongoing maintenance, upgrades, and adaptation as technologies evolve and urban contexts change. Yet governance frameworks emphasize initial deployment rather than long-term management. Institutional arrangements for managing legacy systems when vendors depart or technologies become obsolete remain unclear. Financial provisions for maintenance and upgrades prove inadequate, creating risks of system degradation and stranded assets. Change management processes for updating systems without disrupting critical services are underdeveloped.

Financing

The financing architecture of smart city initiatives across the twelve cities reveals troubling patterns of external dependency, debt accumulation, opacity, and misaligned incentives that raise serious questions about sustainability, sovereignty, and the distribution of benefits and costs. Rather than domestically resourced investments generating returns that fund ongoing development, smart city financing relies heavily on external borrowing that increases debt burdens while locking cities into long-term vendor relationships on unfavourable terms.

External debt financing dominates smart city investments across multiple cities. Nairobi's smart city initiatives have been financed through Korean loans totalling approximately Ksh 7.9 billion for intelligent transport systems and $284 million for Konza Digital Media City, Chinese concessional loans of Ksh 17.5 billion for data centre and smart infrastructure, and Italian financing through UniCredit of $391 million for Konza Phase One infrastructure. Total documented external borrowing exceeds $1.3 billion for Nairobi alone. Senegal's smart city initiatives, including Diamniadio Lake City, rely heavily on Chinese loans and Arab development finance, adding to an external debt burden that has raised sustainability concerns. Egypt's New Administrative Capital, with total costs estimated in tens of billions of dollars, has been financed through multiple external sources including Chinese loans, UAE

investments, and domestic bonds, contributing to debt levels that triggered an IMF program.

This external borrowing pattern creates multiple vulnerabilities. Debt service obligations strain fiscal capacity, redirecting resources from social services and maintenance toward creditor payments. Kenya's external debt stands at approximately seven times the annual budget, with China accounting for roughly 60% of loans, constraining fiscal space for domestically determined priorities. Currency risks accompany foreign-denominated borrowing, as exchange rate movements can dramatically increase local-currency debt burdens. Political risks emerge as creditor nations gain leverage over debtor governments, potentially constraining foreign policy autonomy and creating opportunities for geopolitical coercion.

Opacity surrounding financing arrangements compounds these vulnerabilities. Contract terms for Chinese-financed projects typically include confidentiality clauses preventing public disclosure of loan conditions, interest rates, repayment schedules, and collateral arrangements. This opacity prevents parliamentary oversight, civil society scrutiny, and public debate about whether terms represent value for money. Citizens are asked to service debts whose terms remain unknown, violating basic principles of democratic accountability. Even multilateral financing, while generally more transparent than bilateral arrangements, often fails to provide sufficient detail about conditionalities and expected returns.

Vendor financing arrangements create particularly concerning conflicts of interest and opportunities for corruption. When technology vendors provide or arrange financing for smart city projects, they become both creditors and suppliers, creating incentives to inflate costs and discourage competitive bidding. Export credit agencies from creditor nations promote their domestic technology companies, conflating development finance with industrial policy. Bundled financing-procurement arrangements prevent cities from selecting best-value technology solutions, instead locking them into vendor ecosystems determined by financing sources. These arrangements prioritize vendor interests over municipal needs, undermining value-for-money objectives.

Revenue generation models remain underdeveloped, undermining financial sustainability. Most smart city systems operate as cost centres rather than revenue generators. While digital revenue collection has improved municipal finances in

cities like Accra and Lagos, these gains often fall short of covering smart city investment and operating costs. The expectation that greenfield developments like Konza Technopolis will eventually generate significant economic returns through agglomeration effects and technology sector growth remains speculative, as projects have experienced substantial delays and private sector uptake has lagged projections. Business models for monetizing data generated by smart city systems remain poorly developed, while privacy concerns appropriately constrain exploitative data commercialization.

Maintenance and upgrade financing receives inadequate attention during initial procurement, creating risks of system degradation and functionality loss. Technology systems require ongoing investment in software updates, hardware replacements, capacity expansion, and adaptation to evolving user needs. Yet financing arrangements typically concentrate on capital expenditure for initial deployment, with limited provisions for recurrent costs. When initial vendor warranties expire and maintenance contracts require renewal, cities face locked-in relationships with incumbent vendors who can charge premium prices due to proprietary technologies and switching costs. This dynamic transfers risk from vendors to cities while extracting ongoing rents.

Opportunity costs of smart city investments deserve greater scrutiny. Resources directed toward technology deployments could alternatively fund traditional infrastructure, social services, or human capital development. Do smart traffic signals represent better value than expanded public transportation? Do surveillance systems prevent more crime per dollar than community policing? Do digital platforms improve service delivery more cost-effectively than hiring and training civil servants? These questions rarely receive rigorous analysis, as smart city rhetoric assumes technology provides inherent value regardless of context or comparison with alternatives.

Distributional consequences of financing structures concentrate benefits among vendors, elites, and affluent neighbourhoods while socializing costs across entire populations. International technology companies and their local partners capture contracts and profits. Affluent neighbourhoods receive infrastructure investments that increase property values and quality of life. Middle-class citizens gain access to improved digital services. Meanwhile, poor communities receive minimal direct benefits yet contribute to debt service through taxes on basic goods and services.

This regressive distribution exacerbates existing inequalities, contradicting smart city rhetoric about inclusive development.

Private sector participation, while potentially beneficial, often occurs on terms that privatize profits while socializing risks and costs. Public-private partnerships transfer commercially viable components to private operators—telecommunications infrastructure, data centres, digital payment systems—while leaving unprofitable but socially important functions to government. Private partners frequently negotiate revenue guarantees, minimum return provisions, or other protections that shift risk to government. Tax incentives and subsidies intended to attract private investment reduce potential public revenues. Intellectual property arrangements favour private partners, preventing government from fully utilizing systems it financed.

Multilateral development finance, while offering more favourable terms than commercial borrowing, arrives with conditionalities that constrain policy autonomy. World Bank digital development loans tie financing to specific policy reforms, including regulatory changes that may Favor foreign investors over domestic policy preferences. Conditions requiring open markets, competitive neutrality, and intellectual property protection can prevent governments from pursuing industrial policies supporting domestic technology sectors. Reporting requirements and monitoring obligations create administrative burdens for capacity-constrained governments. While some conditionalities promote good governance, others reflect donor ideological preferences rather than context-specific development needs.

Fiscal implications extend beyond direct debt service. Complementary investments required for smart city systems to function—electricity infrastructure, internet connectivity, digital skills training—create additional fiscal demands. Security and privacy requirements impose ongoing costs for cybersecurity systems, data protection compliance, and regulatory enforcement. Social safety net provisions may become necessary when automation displaces workers or digital service delivery excludes populations unable to access digital platforms. These indirect fiscal implications rarely feature in initial cost-benefit analyses, creating fiscal surprises during implementation.

Ownership

The ownership structures and technology transfer arrangements underlying smart city initiatives across the twelve cities reveal patterns of dependency, foreign control, and vendor lock-in that fundamentally compromise African technological sovereignty and contradict continental aspirations for indigenous digital capacity. Rather than building local capability to manage, maintain, and evolve digital urban infrastructure, current approaches entrench reliance on foreign technology vendors who retain effective control over critical systems while African governments and citizens bear the financial costs and risks.

Foreign Vendor Dominance and Control

Chinese technology firms, particularly Huawei and its surveillance subsidiary Hikvision, dominate smart city implementations across the majority of the twelve cities examined. Nairobi's Safe City surveillance system worth Ksh 17.5 billion was implemented by Huawei, which also supplied the data centre infrastructure for Konza Technopolis. Kampala's Safe City CCTV network was installed by Huawei with approximately 200 cameras equipped with facial recognition capabilities. Lagos has deployed Huawei surveillance infrastructure across the city. Cairo's smart city components rely heavily on Chinese technology providers. Lusaka, Kigali, and other cities have similarly turned to Chinese firms for surveillance and smart city infrastructure. This concentration of implementation responsibility in a single nation's technology ecosystem creates profound dependencies and strategic vulnerabilities.

Beyond Chinese vendors, South Korean firms play significant roles. Samsung Construction and Trading Corporation holds the Ksh 7.9 billion contract for Nairobi's Intelligent Transport System. Korean financing for smart city initiatives in Nairobi, including $284.1 million for Konza Digital Media City, comes bundled with Korean technology and implementation partners. Other cities have engaged Indian, European, and North American technology firms, but the overall pattern remains one of foreign vendor dominance with minimal participation by African technology companies in core system implementation.

The nature of these vendor relationships reflects technology provision rather than genuine partnership. African governmental entities—whether national ministries, municipal authorities, or special purpose development agencies—function primarily as clients purchasing complete technology solutions rather than as co-developers building shared capacity. Vendors arrive with pre-packaged systems

designed for other markets, implement these systems with minimal adaptation to local contexts, provide training limited to basic system operation, and depart leaving governments dependent on vendor support for any substantial maintenance, modification, or troubleshooting.

Ambiguous and Disadvantageous Ownership Arrangements

Data ownership structures remain fundamentally unclear across smart city initiatives, creating risks and uncertainties that favour vendors over governments and citizens. Huawei explicitly defines its role in surveillance systems as technology provider rather than guarantor of privacy rights or custodian of data, placing full operational responsibility on government agencies while simultaneously retaining technical control and access capabilities. The distinction between legal ownership of data and practical control over data becomes critical when governments lack independent capacity to access, extract, analyse, or migrate data without vendor assistance.

Contracts governing smart city projects rarely provide detailed, public clarity on data ownership, access rights, retention obligations, portability provisions, or exit procedures. When contracts remain confidential, as Chinese-financed projects typically require, even governments often have incomplete understanding of their rights and limitations relative to vendor prerogatives. Key questions remain unanswered or ambiguously addressed: Who owns surveillance footage and associated metadata? Who has access rights, under what circumstances, subject to what limitations and oversight? How long can vendors retain copies of data generated by systems they operate? What rights do governments have to data portability if they wish to switch vendors? Under what circumstances might vendors share data with their home governments or other third parties?

Infrastructure ownership similarly reflects ambiguity and disadvantage. While governments typically retain legal ownership of physical infrastructure—cameras, servers, network equipment—they lack ownership of software, algorithms, encryption keys, and technical knowledge necessary to operate systems independently. This separation between physical infrastructure ownership and operational control creates nominal ownership concealing practical dependency. Governments own expensive equipment they cannot independently operate, maintain, or modify, while vendors retain the intellectual property and technical capacity that determines functionality.

The absence of African technology companies in ownership structures represents a profound missed opportunity for building indigenous capacity and ensuring technological solutions align with local contexts and values. When implementation proceeds entirely through foreign vendors, with local participation limited to construction subcontracting and basic service provision, opportunities to build African technology sectors capable of competing regionally and globally are foregone. This pattern contradicts the African Union Digital Transformation Strategy's explicit emphasis on Africa as a producer rather than merely a consumer in the global digital economy, and Research ICT Africa's advocacy for digital sovereignty and locally appropriate solutions developed by African innovators understanding African contexts.

Technology Transfer Deficits

Technology transfer provisions in smart city contracts remain minimal, vague, or entirely absent, representing a critical failure to build long-term local capacity. Genuine technology transfer encompasses multiple dimensions: transfer of technical knowledge enabling local personnel to understand system architectures and operations; transfer of maintenance and troubleshooting capabilities allowing local teams to address problems independently; transfer of modification and upgrade capabilities enabling adaptation to evolving needs without vendor dependency; and transfer of design and development capabilities allowing local teams to eventually create similar systems independently.

Current smart city contracts rarely address these dimensions systematically. Training provisions, when they exist, focus on basic system operation—how to use interfaces, run standard reports, and perform routine tasks—rather than deep technical understanding. Local personnel learn to operate systems as end-users, not to maintain them as technicians or modify them as engineers. When problems arise beyond routine operations, vendor support becomes necessary. When systems require updates or modifications, vendors must be engaged. When expansion or enhancement is desired, the same vendors must be contracted. This pattern ensures ongoing vendor revenue streams while preventing local capacity development.

The distinction between training and technology transfer deserves emphasis. Training creates operators; technology transfer creates engineers. Training enables use of existing systems; technology transfer enables creation of new

systems. Training focuses on specific vendor products; technology transfer develops generalizable capabilities applicable across technology contexts. Current smart city approaches prioritize training while systematically avoiding genuine technology transfer that would reduce vendor leverage and enable competition.

Source code access restrictions exemplify technology transfer limitations. Proprietary systems arrive as "black boxes" where governments can observe inputs and outputs but cannot examine or modify internal operations. Source code—the human-readable instructions that define system behaviour—remains vendor intellectual property, unavailable to government technical teams. This prevents governments from verifying system security, identifying vulnerabilities, customizing functionality, fixing bugs independently, or understanding precisely what data is collected and how it is processed. Open-source alternatives, which would provide full source code access enabling modification and learning, are systematically excluded from procurement in Favor of proprietary solutions maximizing vendor control.

Documentation provided to governments typically covers user operations rather than system architecture, design rationale, development processes, or technical implementation details that would enable local teams to maintain and evolve systems. Comprehensive technical documentation enabling deep understanding remains vendor proprietary information, shared only under restrictive non-disclosure agreements limiting dissemination even within government agencies. This information asymmetry ensures vendors retain technical advantages while governments remain dependent.

Local content requirements, which could mandate vendor use of local engineers, require hiring and training of local personnel, or compel subcontracting to local technology firms, remain minimal or poorly enforced. When requirements exist, vendors typically comply minimally through token hiring or subcontracts for peripheral activities while retaining all core technical work for expatriate teams. Without detailed specifications, monitoring mechanisms, and enforcement consequences, local content provisions fail to achieve technology transfer objectives.

Vendor Lock-In Dynamics and Consequences

Vendor lock-in—situations where switching from one vendor to alternative providers becomes prohibitively expensive, technically complex, or functionally impossible—characterizes smart city implementations across the twelve cities. This lock-in emerges through multiple mechanisms that individually and cumulatively entrench vendor positions while constraining government options and enabling vendors to extract ongoing rents.

Proprietary technologies create technical lock-in by making systems dependent on vendor-specific platforms, interfaces, protocols, and formats incompatible with alternatives. When surveillance systems use proprietary video formats, storage systems, and analysis algorithms, governments cannot simply switch to different camera vendors without replacing entire infrastructure. When traffic management systems rely on proprietary communication protocols between sensors and control centres, adding sensors from alternative vendors becomes impossible without costly integration work that vendors can price prohibitively. When data is stored in proprietary formats, migrating to alternative systems requires expensive and risky conversion processes.

Contractual lock-in emerges through long-term service agreements, maintenance contracts, and financing arrangements that commit governments to extended vendor relationships. Maintenance contracts may run for decades with automatic renewal provisions and punitive early termination penalties. Financing arrangements may tie debt service to continued vendor engagement, creating financial obligations that extend vendor relationships beyond technical necessity. Exclusivity provisions may prevent governments from engaging alternative vendors for related systems, forcing continued reliance on incumbents.

Knowledge lock-in results from vendor monopolization of system expertise. When only vendor personnel fully understand system operations, governments cannot switch providers without losing critical operational knowledge. When vendor training proves deliberately limited to prevent independent capability, governments lack personnel who could manage transitions to alternative systems. When vendor turnover and non-compete-agreements prevent vendor-trained personnel from sharing knowledge with competitors, governments cannot even hire away vendor staff to build internal capacity or support transition to alternatives.

Data lock-in emerges when data accumulates in proprietary formats within vendor-controlled systems. Years or decades of surveillance footage, traffic patterns, service records, and other operational data become valuable assets, but their value depends on accessibility. When data remains in proprietary formats readable only by vendor software, and when data export capabilities are limited, expensive, or deliberately obstructed, governments face difficult choices between continuing vendor relationships to maintain data access or switching vendors and losing access to historical data. Vendors recognize data lock-in power and design systems accordingly.

Financial lock-in results from sunk costs and switching expenses. Governments invest billions in smart city infrastructure, software licenses, training, and integration. Switching to alternative vendors requires writing off these investments while incurring new costs for replacement systems, data migration, retraining, and re-integration. Even when alternative vendors might provide superior functionality or better value over time, upfront switching costs deter change. Vendors understand this dynamic and price initial deployments to win contracts, knowing that lock-in will enable premium pricing for maintenance, upgrades, and expansions.

Network effects create adoption lock-in when system value depends on widespread use. Digital payment systems, digital identity platforms, and other smart city infrastructure become more valuable as adoption increases, creating pressure to standardize on incumbent systems rather than fragmenting across alternatives. Once a system achieves critical mass, switching becomes difficult because users resist migrating to alternatives lacking equivalent network participation. Vendors leverage network effects to entrench positions and deter competition.

The consequences of vendor lock-in prove costly and strategically damaging. Governments lose negotiating leverage once locked in, enabling vendors to increase prices for maintenance, upgrades, and support without competitive pressure. Governments cannot easily respond to vendor performance failures, security vulnerabilities, or changing needs by switching providers. Innovation becomes dependent on vendor roadmaps rather than government priorities or local needs. When vendor interests diverge from government interests—regarding

pricing, feature development, security practices, or data governance—governments lack options to compel alignment.

Lock-in risks intensify when vendors face financial difficulties, strategic reorientation, or geopolitical complications. If vendors exit markets, face sanctions, or experience bankruptcy, governments are left with orphaned systems lacking ongoing support. The sustainability risks of vendor dependency become apparent when external factors beyond government control determine whether critical urban infrastructure continues functioning.

Sovereignty Implications and Strategic Vulnerabilities

The concentration of smart city system ownership and control in foreign vendors creates profound sovereignty implications that extend beyond technical and financial concerns to fundamental questions about African agency in governing critical infrastructure and protecting citizen data. When Chinese firms control surveillance systems monitoring African cities, when Korean firms manage intelligent transport infrastructure, when Indian or European firms operate digital identity platforms, questions arise about whose interests these systems ultimately serve and what constraints African governments face in asserting sovereign authority.

Data sovereignty concerns emerge most acutely with surveillance systems. When Huawei implements Safe City infrastructure, where does surveillance data physically reside? In servers located in African cities but potentially accessible by Huawei technicians in China? In cloud infrastructure that may span multiple jurisdictions? What prevents foreign intelligence services from requesting or demanding access to surveillance data through vendors subject to their jurisdiction? China's National Intelligence Law requires organizations and citizens to "support, assist, and cooperate with state intelligence work," creating legal obligations on Chinese firms that may conflict with host country data sovereignty. African governments operating surveillance systems dependent on Chinese technology firms cannot guarantee that surveillance data remains exclusively under their control.

The cybersecurity vulnerabilities of vendor-dependent infrastructure compound sovereignty concerns. When governments lack independent capacity to audit system security, identify backdoors, or verify that systems operate only as

specified in contracts, they cannot ensure infrastructure serves only sovereign purposes. Security researchers have documented serious vulnerabilities in Hikvision cameras widely deployed across African cities, including remote access vulnerabilities allowing unauthorized control. When African governments lack capacity to identify and remediate such vulnerabilities independently, they depend on vendors for security that vendors may have insufficient incentive to provide.

The geopolitical dimensions of vendor dependency create strategic vulnerabilities. When critical urban infrastructure depends on vendors from potentially adversarial or unreliable foreign states, governments face risks of supply disruptions, support withdrawal, or coercive leverage during diplomatic conflicts. United States restrictions on technology exports to Huawei create uncertainties for African governments dependent on Huawei infrastructure regarding future access to components, software updates, and support. Trade conflicts, sanctions regimes, or geopolitical competition among major powers create risks for African cities whose critical infrastructure depends on vendors caught in these conflicts.

The absence of alternatives to foreign vendors reflects historical underinvestment in African technology sectors and education systems but also represents policy choices. African governments could prioritize procurement from African vendors, even accepting somewhat higher costs or initially inferior functionality to build domestic industries. They could mandate technology transfer and local content requirements aggressively rather than accepting vendor preferences for turnkey implementations. They could invest in education and research institutions building domestic capacity for smart city development. They could collaborate regionally to create sufficient market scale for African technology firms to compete effectively. Current policy choices generally forego these sovereignty-enhancing approaches in Favor of expedient reliance on foreign vendors.

Inclusivity

Smart city initiatives across the twelve cities demonstrate alarming patterns of exclusion that contradict rhetoric about inclusive digital transformation. Rather than bridging divides and ensuring equitable access to services and opportunities, current approaches risk entrenching and amplifying existing inequalities along lines of income, geography, gender, age, disability, and digital literacy. The promise of smart cities for all confronts the reality of smart cities for the connected elite.

Digital infrastructure deployment exhibits stark spatial inequalities that mirror and reinforce socioeconomic divides. Affluent neighbourhoods receive high-speed fibre optic connectivity, ubiquitous 4G and emerging 5G mobile coverage, and dense deployments of smart city sensors and infrastructure. Poor neighbourhoods, particularly informal settlements that house significant proportions of urban populations, experience dramatically inferior connectivity. In Nairobi's Mathare informal settlement, residents face limited reliable internet access despite proximity to the city centre. Lagos's informal settlements lack basic infrastructure, let alone digital connectivity. Johannesburg's spatial apartheid legacy persists in digital infrastructure distribution, with affluent northern suburbs enjoying connectivity that remains aspirational for residents of townships and informal settlements on the periphery.

This infrastructure inequality creates cascading disadvantages. Without connectivity, residents cannot access digital government services, requiring expensive and time-consuming journeys to physical offices. Students cannot participate in online learning, exacerbating educational inequalities exposed during COVID-19 school closures. Entrepreneurs cannot access digital marketplaces or payment systems, constraining informal economy participants to cash-based transactions with limited geographic reach. Job seekers cannot access online opportunities in the platform economy. Citizens cannot engage with e-participation platforms or access information about government decisions affecting their lives.

Affordability barriers exclude large populations even where infrastructure exists. While internet costs have declined, they remain prohibitive for low-income households as a proportion of income. The cost of smartphones capable of running contemporary applications excludes the poorest populations from digital participation. Data costs for sustained internet access consume significant portions of household budgets, forcing difficult choices between connectivity and other essentials. The AU Digital Transformation Strategy's aspiration for internet access at no more than 1 cent USD per megabyte by 2030 remains far from reality in most contexts. Zero-rating arrangements that provide free access to specific platforms create unequal digital experiences while raising net neutrality concerns.

Device access reinforces exclusion. While mobile phone penetration has increased dramatically, smartphone ownership remains limited among low-income

populations. Feature phones incapable of running contemporary applications predominate among the poor, restricting participation to basic services like mobile money while excluding richer digital experiences. The cost of replacing devices as applications demand increasing processing power creates ongoing financial barriers. Shared device use within households, common among the poor, creates privacy concerns and limits individual autonomy. The AU objective of smartphones manufactured in Africa at no more than $100 USD by 2030 has not materialized, with devices typically costing multiples of this target.

Digital literacy gaps limit effective participation even when infrastructure and devices are accessible. Many citizens, particularly older adults, rural-urban migrants, and those with limited formal education, lack skills to navigate digital interfaces, evaluate online information, or protect themselves from digital risks. Training programs reach only small fractions of populations requiring skills development. Interface designs typically assume literacy, numeracy, and familiarity with digital conventions, creating barriers for those lacking these capabilities. Language barriers compound exclusion, as applications typically prioritize major international languages and national lingua francas over minority languages and oral traditions.

Gender dimensions of digital exclusion receive inadequate attention despite well-documented patterns. Women experience lower mobile phone ownership, internet access, and digital literacy than men across African contexts. Social norms in some contexts restrict women's digital access, with male household members controlling devices and digital resources. Safety concerns about online harassment and gender-based violence deter women's digital participation. Digital platforms frequently lack gender-sensitive design, failing to accommodate women's specific needs, time constraints, and security concerns. Smart city initiatives rarely include explicit gender equality objectives or mechanisms to ensure women benefit equitably from investments.

Age-related exclusions affect both young and old. While youth digital adoption rates exceed older cohorts, young people without resources for devices and connectivity remain excluded despite enthusiasm and aptitude. Digital platforms offering economic opportunities frequently impose age restrictions excluding minors, limiting youth economic participation. Older adults experience greatest difficulty adapting to digital service delivery, facing potential exclusion as

traditional in-person services are withdrawn. Age verification systems can create barriers for those lacking official identification documents, disproportionately affecting young people from poor communities where birth registration remains incomplete.

Disability exclusion persists despite assistive technology potential. Digital interfaces rarely follow accessibility guidelines enabling use by persons with visual, hearing, mobility, or cognitive disabilities. Public digital infrastructure like information kiosks and payment terminals typically lack accessibility features. Digital service delivery migration can worsen exclusion for persons with disabilities who previously received in-person assistance. While some cities have implemented accessible features in new developments, retrofitting existing infrastructure and ensuring universal accessibility remains exceptional rather than standard practice.

Education level creates strong digital inclusion gradients. Tertiary-educated populations adopt digital technologies readily and extract maximum value from smart city systems. Secondary-educated populations participate with some constraints. Primary-educated and unschooled populations face the greatest barriers, excluding them from digital economic opportunities and creating dependencies on digitally literate intermediaries who may charge for services that should be freely accessible. This correlation between education and digital inclusion risks creating self-reinforcing cycles where educational advantages translate into digital advantages that further enhance educational and economic opportunities for children of digitally connected parents.

Geographic exclusions extend beyond urban-rural divides to intra-urban inequalities. Even within cities, peripheral neighbourhoods, informal settlements, and lower-income areas experience inferior digital infrastructure and services compared to central business districts and affluent residential zones. This pattern reflects rational commercial decisions by infrastructure providers to prioritize high-return areas but contradicts public policy objectives of universal access. Smart city initiatives concentrate investments in showcase zones—new developments, government districts, tourist areas—rather than distributing benefits equitably across urban space. This creates visible modernity in privileged zones while leaving majority populations in digitally under-served areas.

Ethnic and linguistic minorities face specific exclusion risks. Digital platforms typically prioritize majority languages, creating barriers for minority language speakers. Interface designs may embed cultural assumptions unfamiliar to certain communities. Algorithmic systems trained on majority populations may perform poorly for minorities, creating differential service quality. Surveillance systems raise particular concerns in contexts where ethnic tensions exist, as monitoring capabilities could enable discriminatory policing. Smart city initiatives rarely include explicit minority rights protections or mechanisms to ensure minority communities benefit from investments.

Informal economy participants, who constitute large proportions of urban workforces in African cities, experience systematic exclusion from smart city benefits. Digital registration and licensing requirements can formalize businesses in ways that increase costs and regulatory burdens without commensurate benefits. Smart city systems designed around formal economy assumptions—registered addresses, bank accounts, tax identification numbers—create barriers for informal operators. Digital payment system adoption requirements can exclude customers lacking smartphones or internet access, forcing informal businesses to forgo sales or maintain parallel cash systems at additional cost. Surveillance systems can enhance enforcement against informal traders, using smart city technologies for punitive rather than supportive purposes.

Refugee and displaced populations face particular vulnerabilities. Those lacking official documentation struggle to access digital identity systems that increasingly gate services. Linguistic differences and unfamiliarity with local systems create additional barriers. Poverty and precarity limit device and connectivity access. Surveillance concerns may deter engagement with government digital platforms. Smart city initiatives rarely consider refugee populations' specific needs despite significant urban refugee populations in cities like Nairobi, Kampala, and others.

Accountability

Accountability mechanisms for smart city initiatives across the twelve cities remain severely underdeveloped, creating governance gaps that enable waste, corruption, poor performance, and rights violations to proceed with limited oversight or consequence. The complexity of smart city systems, opacity of contracts, inadequacy of oversight institutions, weakness of participation

mechanisms, and limited access to remedy create an accountability deficit that undermines democratic governance and development effectiveness.

Procurement transparency proves consistently inadequate across smart city initiatives. While most cities have formal procurement regulations requiring competitive bidding and public disclosure, implementation falls far short of these standards. Contracts are often awarded through non-competitive processes justified by urgency, technical complexity, or national security considerations. When competitive processes occur, evaluation criteria favour international vendors with sophisticated marketing capabilities over potentially superior local alternatives. Contract terms remain confidential, preventing public scrutiny of value-for-money, data ownership provisions, technology transfer requirements, or exit conditions.

Chinese-financed smart city projects exemplify procurement opacity. Contracts typically include confidentiality clauses preventing disclosure of terms. Bundled financing-procurement arrangements predetermine technology suppliers, eliminating meaningful competition. Governments receive complete packages—financing, technology, implementation—with limited opportunity to influence specifications or terms. This model accelerates implementation but sacrifices transparency, competition, and alignment with local needs. Citizens learn about smart city projects from media coverage of launch ceremonies rather than participatory planning processes.

Performance accountability remains limited by absence of clear metrics, inadequate monitoring systems, and weak enforcement of contractual obligations. Smart city contracts often lack specific, measurable performance indicators beyond basic functionality requirements. When systems underperform or fail to deliver promised benefits, consequences for vendors prove rare. Contract clauses that should protect government interests—performance bonds, liquidated damages, warranties—are often negotiated away or not enforced. Vendor claims about system capabilities frequently exceed realized performance, yet misrepresentation consequences remain minimal.

Financial accountability suffers from inadequate auditing, limited fiscal transparency, and weak consequences for mismanagement or corruption. While supreme audit institutions exist in most countries, they often lack technical capacity to audit complex technology procurements effectively. When audits

identify irregularities, follow-up remains weak, with parliamentary oversight committees unable to compel corrective action or sanction responsible officials. Corruption in smart city procurement—inflated costs, kickbacks, nepotistic awards—proceeds with limited risk of consequences. Civil society organizations attempting to conduct social audits or budget monitoring face information access barriers and limited channels to influence policy.

Data governance accountability proves particularly weak despite profound implications for rights and democracy. Citizens have minimal visibility into what data smart city systems collect, how it is stored and processed, who has access, how long it is retained, and for what purposes it is used. Data protection laws, where enacted, often exempt government agencies or provide broad national security exceptions. Data Protection Authorities, where they exist, typically lack resources, technical capacity, and political independence to effectively oversee government data practices. Data breaches often go unreported, violating notification requirements that exist on paper but lack enforcement. Data sharing arrangements between government agencies and with private partners remain opaque, creating risks of mission creep and unauthorized uses.

Surveillance accountability mechanisms prove especially deficient given the sensitive nature of monitoring technologies. Most cities deploying surveillance systems lack dedicated legal frameworks governing use, creating ambiguity about permissible purposes, necessary authorization procedures, oversight mechanisms, and redress channels. Surveillance policies, where they exist, grant broad discretion to security agencies with minimal oversight. Independent oversight bodies capable of accessing systems, reviewing footage, and investigating complaints rarely exist. Statistics on surveillance use—footage reviewed, investigations initiated, arrests made—remain undisclosed, preventing assessment of necessity, proportionality, or effectiveness.

The lack of impact assessments prior to smart city deployments represents a significant accountability failure. Privacy impact assessments, human rights impact assessments, environmental impact assessments, and social impact assessments could identify risks and inform mitigation measures before irreversible commitments are made. Yet systematic impact assessment remains exceptional rather than standard practice. When assessments occur, they often amount to box-checking exercises rather than genuine risk analyses informing project design.

Public consultation during impact assessments, where it occurs, proves superficial and fails to meaningfully incorporate community input.

Vendor accountability remains limited by asymmetric power relationships and lock-in dynamics. Once smart city systems are deployed, switching costs create vendor leverage to renegotiate terms, increase prices, or resist accountability demands. Proprietary technologies prevent governments from independently maintaining, modifying, or auditing systems, creating information asymmetries where vendors possess comprehensive system knowledge while governments remain partially blind to system operations. Vendor liability limitations negotiated into contracts leave governments bearing risks from system failures or security breaches. When systems underperform or become obsolete, governments often lack exit strategies or alternatives, forcing continued investment in failing systems.

Access to information remains restricted despite rhetorical commitments to transparency. Freedom of information laws exist in several countries but implementation proves uneven, with frequent use of exemptions, lengthy delays, and informal resistance to disclosure. Smart city data that should be public—procurement records, performance statistics, impact evaluations—remains difficult or impossible for citizens to access. Open data portals, where they exist, often contain limited data of poor quality, irregular updates, and formats hindering analysis. Proactive disclosure obligations remain minimal, placing burden on citizens to request information rather than requiring government transparency by default.

Participation mechanisms prove weak or absent across most smart city initiatives. While rhetorical commitments to multi-stakeholder governance appear in policy documents, meaningful participation in planning, oversight, and evaluation remains rare. Consultation processes, when they occur, take place after key decisions are made, reducing participation to rubber-stamping predetermined outcomes. Marginalized communities most affected by smart city deployments—informal settlement residents, street vendors, youth—have minimal voice in processes dominated by officials and vendors. Digital participation platforms intended to broaden engagement paradoxically exclude those lacking digital access, reinforcing rather than ameliorating exclusion.

Civil society oversight capacity remains constrained by limited technical capacity, information access barriers, funding constraints, and intimidation risks.

Organizations attempting to monitor smart city initiatives struggle to access information about contracts, procurement processes, and performance. Technical complexity creates barriers to effective analysis, with limited availability of independent technical expertise. Funding constraints limit sustained engagement, as donor priorities shift and government grants create dependency. Civil society critics of smart city initiatives sometimes face harassment, surveillance, or legal action, chilling criticism and investigation.

Redress mechanisms for citizens harmed by smart city systems prove inadequate. When digital platforms exclude citizens from services, when surveillance systems misidentify individuals, when data breaches expose personal information, affected citizens face limited channels for remedy. Complaint mechanisms often involve approaching the very agencies responsible for problems, creating conflicts of interest. Administrative law procedures prove inaccessible to most citizens due to cost, complexity, and limited legal aid. Courts lack technical expertise to adjudicate technology-related disputes effectively. Alternative dispute resolution mechanisms remain undeveloped for smart city grievances.

Parliamentary oversight proves weak despite formal powers. Legislators typically lack technical literacy to effectively scrutinize smart city proposals and performance. Parliamentary committees reviewing smart city budgets and policies often lack capacity for rigorous analysis, relying on information from government officials and vendors with interests in securing approvals. Opposition legislators attempting oversight face information access barriers and limited ability to compel witnesses or documents. Parliamentary proceedings may be manipulated through rushed approvals, limited debate time, or intimidation of critical voices.

International accountability mechanisms remain underdeveloped for smart city initiatives despite international financing, technology flows, and rights implications. When foreign vendors implement systems in African cities, limited mechanisms exist to hold them accountable for failures or rights violations in their home jurisdictions. International development finance institutions have accountability mechanisms theoretically accessible to affected communities, but awareness remains limited and procedures prove complex and lengthy. Regional human rights courts could potentially review smart city rights impacts, but cases remain rare and remedies uncertain.

Media oversight, potentially a strong accountability mechanism, faces constraints that limit effectiveness. Technical complexity makes smart city stories difficult for generalist journalists to investigate and explain. Official sources dominate coverage, with limited independent analysis or affected community voices. Commercial pressures discourage investigative reporting on powerful vendors who purchase advertising. Government pressure, direct and subtle, discourages critical coverage. When strong investigative journalism exposing smart city failures or corruption does occur, follow-up to compel accountability proves limited.

The temporal dimension of accountability deserves emphasis. Smart city systems operate over decades, yet political cycles create incentives for officials to prioritize short-term launches over long-term performance. When systems fail or underperform years after deployment, officials who made initial decisions have often moved on, creating diffuse accountability. Long-term consequences—debt burdens, technological lock-in, rights implications—manifest after political attention shifts elsewhere. Institutional memory proves limited, with lessons from previous failed initiatives often not informing subsequent decisions.

Surveillance

Surveillance systems represent the most troubling dimension of smart city initiatives across the twelve cities, with extensive deployment of monitoring technologies proceeding with minimal legal frameworks, inadequate oversight mechanisms, limited transparency, and profound implications for privacy, freedom of expression, freedom of association, and political participation. The normalization of comprehensive monitoring under "smart city" and "public safety" branding obscures the fundamental transformation of state-society relationships that ubiquitous surveillance enables.

Scale and sophistication of surveillance systems deployed across these cities warrant careful documentation. Nairobi's Safe City network comprises approximately 2,000 CCTV cameras with facial recognition capabilities, operating 24/7 with footage stored in cloud systems managed by Huawei. Johannesburg has deployed extensive CCTV networks across the city, with particular density in commercial districts. Cairo's surveillance infrastructure has expanded dramatically, with thousands of cameras monitoring public spaces and integration with broader national security monitoring systems. Lagos has implemented

surveillance systems justified by security concerns, with installations expanding into markets, transportation hubs, and major arteries. Kampala, Kigali, Rabat, Tunis, Lusaka, Dakar, Accra, and Abidjan have all deployed or announced plans for surveillance systems marketed as Smart City or Safe City initiatives, typically implemented by Chinese technology firms.

The technological capabilities of these systems extend well beyond traditional CCTV. Facial recognition enables identification of specific individuals from footage, creating possibilities for comprehensive tracking of movements and associations across urban space. License plate recognition tracks vehicle movements, revealing travel patterns and potentially location histories. Integration with other databases—national ID systems, vehicle registration, criminal records, mobile phone records—enables correlation of surveillance data with identity and biographical information. Some systems incorporate behavioural analysis algorithms claiming to predict suspicious activities or identify anomalies worthy of investigation. Video analytics enable retrospective searches of footage archives for specific individuals, vehicles, or behaviours across time and space.

Huawei's dominant role in deploying surveillance infrastructure across African cities raises sovereignty and security concerns beyond surveillance implications for citizens. Huawei's "Safe City" platform has been implemented in Nairobi, Kampala, and other cities, with Hikvision cameras providing hardware. This creates dependencies on Chinese technology firms for critical security infrastructure while raising questions about data access, backdoors, and potential use of African surveillance data by Chinese intelligence services or for training algorithms used globally. When surveillance infrastructure depends on foreign technology, host governments cannot fully control access or prevent external exploitation, creating national security vulnerabilities even as governments deploy these systems ostensibly to enhance security.

The legal frameworks governing surveillance deployment and use remain grossly inadequate across these twelve cities. Most countries lack dedicated surveillance legislation establishing permissible purposes, authorization requirements, oversight mechanisms, data retention limits, and redress procedures. When security legislation exists, it typically grants broad surveillance powers with national security justifications, minimal judicial oversight, and limited transparency. Kenya's Prevention of Terrorism Act and National Intelligence

Service Act provide expansive surveillance powers with inadequate safeguards. South Africa's Regulation of Interception of Communications and Provision of Communication-Related Information Act provides some procedural safeguards but implementation oversight remains limited. Many countries operate surveillance systems in legal grey zones where authority claims rely on general security mandates rather than specific legislative authorization.

Data protection frameworks, where they exist, often contain broad exemptions for security and law enforcement that effectively exclude surveillance from privacy protections. Kenya's Data Protection Act 2019 establishes important principles but exempts national security activities. South Africa's Protection of Personal Information Act contains similar exemptions. Countries that have ratified the AU Malabo Convention commit to data protection principles but implementation lags far behind formal ratification. The result is a regulatory gap where comprehensive surveillance proceeds with minimal legal constraint or oversight.

Oversight mechanisms prove profoundly inadequate relative to surveillance capabilities and implications. Independent oversight bodies capable of accessing surveillance systems, reviewing footage and use logs, investigating complaints, and compelling corrective action rarely exist. Parliamentary oversight committees lack technical capacity, access rights, and time to effectively monitor surveillance use. Data Protection Authorities, where they exist, often lack jurisdiction over security agencies or capacity to audit surveillance practices effectively. Judicial oversight proves limited, with surveillance authorization decisions often made by executive officials rather than judges, and minimal transparency enabling assessment of decision quality or appeal rights.

Transparency regarding surveillance systems remains minimal, violating principles essential for democratic accountability. Statistics on surveillance camera numbers and locations, footage review frequency, investigation initiation rates, evidentiary use in criminal proceedings, retention periods, data sharing arrangements, and costs remain largely undisclosed. Procurement contracts remain confidential, preventing public scrutiny of system capabilities, data ownership, access provisions, and vendor responsibilities. Operational policies, if they exist, remain classified on national security grounds. Citizens lack information about whether they are being surveilled, by what systems, for what purposes, with

what authorizations, under what limitations, and with what redress channels if surveillance is used improperly.

Purpose limitation principles essential for constraining surveillance receive minimal attention. Surveillance systems deployed ostensibly for traffic management or public safety can easily be repurposed for political monitoring, protest surveillance, or routine tracking of movements and associations. Without legal prohibitions on purpose expansion and oversight mechanisms to detect and prevent unauthorized uses, mission creep becomes inevitable. Traffic cameras become surveillance cameras. Public safety systems become political control systems. Technical capabilities enable uses that exceed initial justifications, and inadequate governance fails to constrain opportunistic expansion.

The chilling effects of surveillance on freedom of expression and association deserve emphasis. When citizens know or suspect that movements, associations, and communications are monitored, self-censorship increases. Political organizing becomes riskier as surveillance enables identification of activists, mapping of networks, and anticipatory disruption of mobilization. Journalists investigating government actions face heightened risks. Civil society organizations experience constrained space. Minority communities subjected to disproportionate surveillance alter behaviours to avoid triggering algorithmic suspicion indicators. These behavioural changes represent harms even absent direct persecution, as surveillance shapes behaviour through internalized anticipation of monitoring.

The use of surveillance for political purposes, while difficult to document due to opacity, emerges regularly in African contexts. Surveillance infrastructure has been used to identify and arrest opposition activists, monitor journalists, track civil society organizations, and enable pre-emptive action against planned protests. While governments justify surveillance as security measures, practice reveals elastic security definitions that encompass political dissent, peaceful protest, and investigative journalism. The same infrastructure deployed for crime prevention becomes available for political control, and inadequate oversight mechanisms fail to prevent political use.

Differential surveillance of marginalized communities compounds existing discrimination. Evidence from multiple contexts suggests that surveillance cameras concentrate in poor neighbourhoods and informal settlements rather than affluent areas, inverting the crime distribution that presumably justifies

surveillance. Racial and ethnic profiling shapes surveillance targeting, with minority communities subjected to disproportionate monitoring. This pattern criminalizes poverty and marginalizes identity rather than pursuing evidence-based security provision. It reflects surveillance serving social control objectives rather than or beyond public safety.

Algorithmic bias in facial recognition and behavioural analysis systems creates discriminatory outcomes. Facial recognition algorithms trained primarily on lighter-skinned faces perform poorly on darker-skinned individuals, creating higher false positive rates that could lead to mistaken identifications and wrongful accusations. Behavioural analysis algorithms trained on biased data reproduce and amplify human prejudices, flagging behaviours common in marginalized communities as suspicious. These technical failures create discriminatory surveillance impacts that violate equality principles and compound existing injustices.

Data retention practices enable retrospective surveillance with profound implications for freedom of assembly and association. When surveillance footage and associated metadata are retained for extended periods, authorities can retrospectively identify protesters, track activists over time, and map organizational networks after protests or political activities. This retrospective capability means surveillance harms extend beyond immediate monitoring to create permanent records available for future use. Without retention limits and automated deletion requirements, surveillance creates permanent databases of movements and associations available for political or security uses.

Data sharing arrangements compound surveillance risks. When surveillance systems interface with other databases—national ID, mobile phone records, social media, financial transactions—correlation enables comprehensive profiling of individuals. Integration of surveillance footage with other data sources enables identification of individuals in contexts where they might otherwise be anonymous. Cross-border data sharing arrangements can enable authoritarian governments to track citizens abroad or facilitate persecution of refugees and dissidents. Inadequate data governance means these integration and sharing capabilities proceed with minimal constraint or oversight.

Private sector involvement in surveillance creates additional accountability challenges. When private security firms or telecommunications companies operate

surveillance systems or provide access to data, mixed public-private accountability complicates oversight. Private firms may resist disclosure of practices based on commercial confidentiality claims. Contractual relationships between government and private operators may diffuse accountability. Profit incentives may drive expansion of surveillance beyond public interest justifications. International private sector surveillance vendors may resist host country legal jurisdiction.

The resilience of surveillance infrastructure beyond the regimes that deploy it deserves consideration. Surveillance systems, once installed, remain available to successor governments regardless of political orientation. Authoritarian regimes installing comprehensive surveillance create infrastructure that can be exploited by future even less rights-respecting regimes. Democratic transitions do not automatically eliminate surveillance capabilities built under authoritarian predecessors. This path-dependency means surveillance deployment decisions have consequences extending far beyond current governments.

Normalization of surveillance through gradual expansion and positive framing represents a deliberate strategy to reduce resistance. Initially, cameras appear in high-crime areas or major intersections with public safety justifications that attract broad support. Gradually, coverage expands with minimal public discussion. Marketing emphasizes safety, efficiency, and modernity rather than monitoring and control. Citizens become accustomed to camera presence. By the time surveillance becomes comprehensive, resistance potential has diminished and reversal becomes politically difficult. This gradual normalization strategy consciously exploits cognitive biases and limited public attention.

The cybersecurity vulnerabilities of surveillance systems create risks beyond privacy concerns. Insecure surveillance networks can be hacked by criminals, foreign intelligence services, or malicious actors, enabling unauthorized access to footage and potentially real-time monitoring without government knowledge. Researchers have documented serious vulnerabilities in Hikvision cameras widely deployed in African cities. Compromised surveillance systems could facilitate theft, stalking, extortion, or surveillance by actors other than deploying governments. These security weaknesses undermine the security objectives ostensibly justifying surveillance while amplifying privacy risks.

Conclusion

This analysis has revealed a complex landscape where genuine benefits coexist with profound governance challenges, financial sustainability concerns, inclusivity deficits, accountability gaps, and surveillance risks. As these initiatives proliferate and deepen, their alignment with continental frameworks established through African Union instruments becomes crucial for ensuring that digital transformation serves African development priorities rather than vendor interests, reinforces democratic governance rather than authoritarian control, and promotes inclusion rather than exclusion. Four key AU instruments provide essential lenses through which to evaluate current trajectories and chart necessary course corrections: the AU Data Policy Framework, the AU Digital Transformation Strategy, the Continental AI Strategy, and the African Digital Compact. Synthesizing across these four instruments, several cross-cutting themes emerge that should guide smart city course corrections.

First, the emphasis on sovereignty and African ownership conflicts fundamentally with vendor-dependent approaches that concentrate control in foreign firms. Moving toward the AU vision requires prioritizing open standards, open-source alternatives, technology transfer requirements, local capacity building, and data sovereignty provisions in procurement frameworks. This represents not technological nationalism but rather strategic insistence on African agency in governing digital transformation.

Second, the emphasis on harmonization, interoperability, and continental integration identifies fragmentation as a critical limitation of current approaches. Creating the digital single market that AU frameworks envision requires institutional innovations for standards development, mutual recognition agreements, data flow protocols, and coordinated procurement that current governance arrangements lack. The African Union Commission, Regional Economic Communities, and pan-African technical institutions have responsibilities to facilitate this coordination that remain substantially unfulfilled.

Third, the emphasis on inclusion, equity, and leaving no one behind provides unambiguous standards against which current smart city approaches demonstrate systematic failure. Addressing this requires not modest incremental adjustments but fundamental reorientation prioritizing connectivity for underserved areas, affordable access, device subsidies for the poor, digital literacy programs at scale,

accessible design, and active efforts to ensure women, youth, persons with disabilities, and marginalized communities benefit equitably.

Fourth, the emphasis on transparency, accountability, and multi-stakeholder governance demands governance reforms far beyond current practice. This requires legal frameworks mandating disclosure, independent oversight institutions with resources and authority, participatory mechanisms with genuine influence over decisions, civil society capacity building, and cultural change within governments toward openness rather than secrecy.

Fifth, the emphasis on data governance, privacy protection, and digital rights requires robust regulatory frameworks, independent Data Protection Authorities, cybersecurity capacity, and legal restrictions on surveillance with meaningful oversight. The surveillance expansion documented in this analysis proceeds in directions opposite to AU framework principles, demanding fundamental reassessment of whether comprehensive monitoring serves development or undermines the rights, freedoms, and democratic governance that sustainable development requires.

The African Union frameworks represent not merely aspirational statements but rather politically endorsed continental commitments reflecting African leaders' collective determination about digital transformation pathways. The dissonance between framework principles and smart city implementation reality suggests that political commitments have not translated into changed practice, vendor influence exceeds policy framework influence, and implementation gaps reflect insufficient prioritization, capacity, or political will. Closing these implementation gaps requires moving beyond policy articulation to enforcement mechanisms, capacity building, resource allocation, and accountability for alignment with continental frameworks.

The path forward requires acknowledging that smart city benefits documented in this analysis are real but insufficient to justify current approaches' governance failures, financial dependencies, exclusionary patterns, accountability deficits, and surveillance expansion. A fundamentally different approach, aligned with African Union frameworks and genuinely serving African development priorities, must prioritize governance before technology, rights alongside efficiency, inclusion as prerequisite for legitimacy, transparency as foundation for accountability, and African sovereignty over digital transformation pathways. The question is not

whether African cities should pursue digital transformation—they must and will—but whether that transformation serves African peoples and their rights, or vendor profits and state surveillance capacity..

Sources and Methodology

Case Studies

The twelve case studies are available here:

- **Abidjan**
- **Accra**
- **Cairo**
- **Dakar**
- **Johannesburg**
- **Kampala**
- **Kigali**
- **Lagos**
- **Lusaka**
- **Nairobi**
- **Rabat**
- **Tunis**

Each city analysis draws on news sources, academic literature, government documents, and data collection spanning ownership, finance, technology, governance, surveillance, performance, social impact, environmental impact, economic benefit, and systems integration. These sources can be found in the annexes to each case study. (The separation of the sources from the main body of text is not ideal but was the result of technical issues instructing the report writing engine.)

Annotated Bibliography

An annotated bibliography of over 100 academic papers and policy briefs

Methodology

The full methodology, outlining the design of the project through incremental iterative interactions with Perplexity, can be found here.